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Methods — Tier-1 Static Synthetic DCE

Working methods document for the Tier-1 static task: **predict a single post-contrast DCE volume from three non-contrast modalities (T2w, DWI, ADC)**. Everything is conditional; inputs are stacked as a 3-channel volume and the model emits a 1-channel DCE volume in $[-1, 1]$.

This file records (a) the UCSF cohort and how it is constructed, and (b) the pipeline defects found and corrected during the 2026-07 audit, with an explicit note on which corrections are reflected in the current results and which are still outstanding. Status as of **2026-08-06**.

1. Data

1.1 Source

Single-center UCSF prostate MRI, delivered on two secured mounts:

mount	contents
CX000018_DS2/.../UCSF_data/registered/<pid>	T2W, ADC_to_T2W, DWI_b*_to_T2W, prostate_mask, prostate_zones — 5962 patients

mount	contents
CX000018_DS3/.../UCSF_data/registered/<pid>/DCE_4D	DCE_4D_to_T2W.nii.gz (4D, phases stacked) + dce_times.json; also pregad_4D_to_T2W.nii.gz + pregad_times.json

All volumes are pre-registered into T2w space by the data provider. `dce_times.json` gives per-phase acquisition times (`{"case", "n_phases", "time_source", "phases": [{"idx", "rel_time_s", ...}]}`), with `idx 0` the pre-contrast phase at `t = 0`.

The pre-contrast T1 (pregad) is present on disk and is **not** currently used; it is available as a 4th conditioning channel or for an enhancement-residual target (`--use-pregad`).

1.2 Target-phase selection: by time, not by intensity

Acquisition protocols are heterogeneous — `n_phases` observed at 26/35/44/70 with cadence 1.7–13 s — so a fixed **phase index** does not correspond to a fixed post-injection time across patients. We therefore select the target phase by **elapsed time**.

The enhancement curve was measured directly (mask-mean signal vs. the pre-contrast phase): flat at $\sim 1.0\times$ until ~ 45 s, steep wash-in from 55–90 s ($1.12\times \rightarrow 2.00\times$), then a **plateau of 2.1–2.5 $\times$ from ~ 90 s out past 300 s**. There is no sharp peak. Intensity `argmax` therefore lands at ~ 280 s on plateau noise and drifts into washout, which is why “peak contrast” is the wrong selection criterion for this cohort.

`--dce-target-time 120` (default) sits inside the stable plateau — past the timing-sensitive wash-in, before washout — and recovers $\sim 92\%$ of maximum enhancement. Realized selection times cluster tightly at 104–125 s.

Two protocol complications are handled explicitly:

- **Interleaved series.** Some studies stack k volumes per timepoint (duplicate timestamps, e.g. Dixon water/fat), so a plain index alternates between sub-series and can land on the non-enhancing one. Detected via repeated `rel_time_s`; the phase is re-picked within the timepoint, keeping the sub-series that actually enhances relative to its own `t = 0`. 27 cases affected.
- **Corrupt timing.** Some `dce_times.json` carry corrupt trailing values (e.g. 67989 s). Only the clean monotonic prefix is trusted; when the time-selected phase still looks weak ($< 1.5\times$), the phase is re-selected from the enhancement curve at **plateau onset** (first phase $\geq 90\%$ of the curve maximum). 155 cases rescued.

1.3 Staging

Each 4D DCE series is ~ 1 GB. Training directly against them re-reads the full 4D volume per sample per epoch (I/O-bound) and depends on a mount that is intermittently absent on compute nodes. `data/stage_ucsf.py` reads each 4D **once**, writes the time-selected phase as a 3D `DCE_to_T2W.nii.gz`, and co-locates T2w/ADC/DWI/mask/zones into a single flat directory per patient with a `stage_meta.json` sidecar (selected index, realized time, enhancement ratios, DWI source filename, interleave factor).

Staging is resumable and shards across a SLURM array (`--shard/--num-shards`). The staged target is verified **bit-identical** to the raw-4D path.

1.4 Cohort construction

Quality filters are applied at two points. The enhancement filter runs **at staging** so the tree is correct by construction; the remaining two are load-time flags.

stage	criterion	remaining
patients with DCE present	—	5439
successfully staged	inputs resolvable, mask matches DCE grid	4975
MIN_ENH=1.5 at staging	max enhancement over the whole series < 1.5×	4246
--dwi-min-bvalue 600	DWI resolves to b < 600	3684
--require-qc	no enhancement QC recorded	3667

- **Enhancement QC** filters on the **maximum over the whole series**, not the selected phase — only the max distinguishes “never enhanced” (drop) from “we picked the wrong phase” (recoverable by re-selection). ~12–13% of patients never enhance at any timepoint; this is a genuine cohort property and belongs in the paper’s data section. 17 further cases have no QC value at all (enhancement is only measurable when the mask matches the DCE grid and is non-empty), so they were staged unverified and are dropped.
- **DWI b-value.** UCSF DWI spans b50–b1400: b1000 (3485), b50 (562), b1400 (135), b1000_synth (29), b1400_synth (13), b0800 (7), b0600 (7), b1350 (5), b600 (2), b1200_synth (1). The 562 b50 acquisitions carry almost no diffusion weighting and read as anatomical/T2-like — a **different contrast** in the same input channel as the b1000 the majority receive. --dwi-min-bvalue 600 enforces channel consistency at the cost of 13% of the data; this is a clean one-flag ablation worth reporting. Note also that **43 cases use computed (_synth_) high-b images rather than acquired ones.**

1.5 Intensity normalization

Single center, so there is no cross-scanner distribution to align and Nyul landmark harmonization is not used; normalization is per-image.

- **DCE** -> --dce-norm robust --dce-robust-k 1 (both must be passed explicitly; the CLI defaults are percentile and k=2.0). The robust scheme keys to the body-tissue median ± k-spread. Measured within-prostate target std over 12 staged cases: k=3 -> 0.073, k=2 -> 0.109, **k=1 -> 0.218**. Bao’s k=3 was tuned by a *cross-hospital alignment* probe that does not apply to a single-center cohort, and it leaves the ROI target nearly flat — the same degeneracy that made held-out metrics meaningless previously, since roi_pearson/ssim_roi become noise on a near-uniform target. Cost of k=1: the contrast-filled bladder saturates at +1, irrelevant for a prostate task.
- **T2w** -> percentile normalization. **ADC** -> fixed physical clip (preserves absolute mm²/s). **DWI** -> per-image foreground z-score.

1.6 Split

Single center, so the split is **patient-level** (--ucsf-test-frac, default 0.15) with a held-out validation split for checkpoint selection. Splits are seeded. ID prefixes were checked as a possible acquisition-batch marker and rejected — 8ii* spans 512, 704, 736 and 1024 matrices alike, so the prefix carries no protocol information and no stratification on it is warranted.

1.7 Geometry — OPEN ISSUE

--spatial-size currently specifies a **pixel count**, and is applied as a center crop/pad on each case’s native grid (--reference dce). UCSF in-plane resolution is **not** uniform: matrices of 256/512/672/704/736/1024 at 0.176–0.781 mm. A fixed 256-pixel crop therefore covers a variable physical field of view (full cohort, n = 4246):

256-px crop covers	cases	share
< 80 mm	734	17%
80–110 mm (intended framing)	3346	79%
> 110 mm	164	4%

An **8× spread in anatomical scale** (40–320 mm). At the low end, 172 cases crop to 45 mm — *smaller than a prostate* — so the crop sits entirely inside the gland: the prostate mask fills the frame, there is no surrounding anatomy, and ROI metrics degenerate to whole-image metrics for those cases.

The through-plane axis carries the same defect at smaller scale: slice thickness is 3.00 mm for 4151/4246 cases (97.8%), but 44 cases are acquired at 0.70–0.80 mm, so a 32-slice crop covers a **25.6 mm slab** for them versus 96 mm for everyone else.

The fix is to resample to a common physical grid before cropping (`--reference iso --iso-spacing 0.35 0.35 3.0`), which makes a 32×256×256 crop cover 90 × 90 × 96 mm for **every** patient. Validated against the cohort:

- the dominant 512 @ 0.352 mm in-plane group and the 97.8% at 3.00 mm slice thickness are left effectively unresampled, so the change is a no-op for the bulk of the data and acts only on the outliers it is meant to correct;
- the iso grid anchors on T2w, and **T2w and DCE share an identical grid in all 4246 cases**, so the resample introduces no crop-center shift relative to the current `--reference dce` behaviour.

`--iso-spacing` was added to the CLI on 2026-07-30 (it did not previously exist, so `--reference iso` would have silently used a 1 mm isotropic grid — a 256 mm field of view with 3–4× z-upsampling).

This is not yet reflected in any trained model. See §6.

2. Preprocessing

All modalities are resampled onto one reference grid so channels are voxel-aligned and stackable (linear interpolation for intensities, nearest-neighbour for mask/zones), then intensity-normalized to $[-1, 1]$ and center-cropped/padded to `--spatial-size` (order D H W; SimpleITK image spacing is (x, y, z)).

Each sample is `{"cond": (3,D,H,W), "target": (1,D,H,W), "mask": (1,D,H,W), "id": str}`. The 2D track wraps the 3D dataset and emits axial slices.

3. Models

All model families expose the same contract: the trainer returns `(model, gen)` where `gen(cond) -> DCE prediction in [-1, 1]`, so evaluation is model-agnostic.

3D. Conditional GAN (pix2pix-style U-Net generator with full skips, hinge loss + reconstruction, projection discriminator); latent DDPM (ϵ -prediction, ancestral + DDIM samplers); latent flow matching. Both LDMs share a frozen first stage — either a 3D AutoencoderKL3D trained on target DCE volumes, or the frozen **MedVAE** foundation VAE (`--first-stage medvae`) — with the conditioning volume downsampled to the latent grid.

2D. gan2d (pix2pix), flow2d (pixel-space CFM), and MedVAE-latent flow.

Flow-matching details. Linear/rectified interpolant with $t = 0 \rightarrow$ data and $t = 1 \rightarrow$ source (reversed relative to much of the literature):

- path: $z_t = (1 - (1 - \sigma_{\min}) \cdot t) \cdot z_0 + t \cdot \text{noise}$
- target velocity: $v = \text{noise} - (1 - \sigma_{\min}) \cdot z_0$, with $\sigma_{\min} = 0$ by default

$t \sim U(0, 1)$ is sampled per-example, so the network learns the velocity field across the whole interpolation, not just the endpoints. `--flow-source` defaults to **noise**; `t2w` is available but has not been used in any run to date. Solvers: 3D uses Heun (2nd-order) by default with Euler available; **2D is Euler-only**.

An optional FlowMI-style **image-space anchor** (`--anchor-weight`) decodes the predicted clean latent $z_0_{\text{hat}} = z_t - t \cdot v$ — an identity that holds exactly for any `sigma_min` — and supervises it with the image-space criterion. `--anchor-t-max` restricts this to low-noise steps, where z_0_{hat} is sharp, to control cost.

4. Loss

$\text{CustomLoss} = l1_weight \cdot L1 + ssim_weight \cdot (1 - \text{SSIM3D}) + \text{perceptual_weight} \cdot \text{Perceptual3D}$, reused as both the GAN generator’s reconstruction term and the VAE’s.

Because the prostate is ~1% of the volume, an unweighted loss is dominated by background and padding. With a mask and `--roi-weight > 1` (default 10), ROI voxels count `roi_weight`× more in L1 and an additional $(1 - \text{SSIM}_{\text{roi}})$ term is added; zone weights (`--tz-weight`, `--pz-weight`) further emphasize transition/peripheral zones. With no mask or `roi_weight ≤ 1` the loss is byte-for-byte the unweighted original.

The perceptual term uses a frozen 3D ResNet (MedicalNet/Med3D).

Note on the LDMs: their diffusion objective is latent-space MSE with no clean ROI mapping, so ROI emphasis reaches them only through the VAE reconstruction stage (or through the image-space anchor, when enabled).

5. Evaluation

Per-volume **SSIM / PSNR / MAE** plus ROI counterparts **SSIM_roi / PSNR_roi / MAE_roi** (computed only where a non-empty mask exists). The global-vs-ROI gap is the intended signal: global metrics are ~99% background, so a large gap means the gland is poorly reconstructed behind good-looking global numbers.

Beyond fidelity, two families:

- **Faithfulness** — `roi_pearson` (within-gland voxelwise co-localization, the hard metric and current ceiling) and `p75_corr` (cross-patient enhancement tracking).
- **Realism** — `roi_var_ratio` (heterogeneity, 1 = match), `roi_grad_ratio` (spatial-frequency/detail, 1 = match), `roi_w1` (ROI intensity-histogram distance), and FID over 2D axial slices. `realism_score` combines the first three; FID is excluded as too unstable at these sample sizes.

`selection_score` supports checkpoint selection on `ssim_roi`, `roi_pearson`, or a balanced objective — realistic **and** faithful, since optimizing pure realism rewards hallucination.

6. Pipeline corrections

Systematic audit, 2026-07. Grouped by what each defect corrupts. Every one was **silent** — no crash, no warning, plausible-looking output.

6.1 Data-construction defects

#	defect	effect	status
D1	DWI resolved against a hardcoded stem list	~1000 patients scanned at b400/b800/b1500/b2000 dropped as “missing inputs”; now globbed with nearest-b preference	fixed, in current cohort
D2	Corrupt timestamps after phase 0 -> select_phase_by_time returned idx 0	shipped the pre-contrast volume as the prediction target for 155 cases — i.e. trained to predict a non-enhanced image	fixed (curve re-selection)
D3	_interleave_factor divided by unique timestamps	null timings looked like $k = n_phases$, masking D2	fixed (requires complete times, capped at $k=4$)
D4	--overwrite re-stage without MIN_ENH	silently re-created the 729 pruned non-enhancers ; a run trained on 4975 before it was noticed	fixed by filtering at staging
D5	_dwi_bval read the staged (renamed) DWI filename	staging renames to canonical DWI_to_T2W.nii.gz, so every case scored $b = -1 < 600$ and every case was dropped — num_samples = 0	fixed (reads dwi_src from stage_meta.json; empty split now raises)
D6	--report wrote to stage_summary.json	clobbered the staging run’s skip records	fixed (stage_report.json)

D2 and D4 are the consequential ones: both put wrong *targets* into training.

6.2 Training defects

#	defect	effect	status
T1	GAN conditioning pooled across the batch	the generator’s condition was contaminated by other patients in the batch	fixed; 3D GAN required retraining
T2	Sampling unseeded in GAN, 2D flow, 3D flow and DDPM	validation integrated from different noise each epoch, so the val curve mixed model improvement with sampling variance — checkpoint selection was partly selecting on noise	fixed (seeded samplers)

6.3 Measurement defects

These corrupt reported numbers but not weights — affected runs can be re-scored from checkpoints rather than retrained.

#	defect	effect	status
M1	ROI scalars pooled over the batch before reduction (roi_radiomics, zone_metrics, eval_metrics)	catastrophic inflation: on pure noise this reports roi_pearson +0.977 where the truth is +0.020 , because cross-patient brightness differences masquerade as within-gland correlation	fixed — all ROI scalars computed per sample, then averaged
M2	2D ROI mask rank mismatch	broadcast to a (B, B, ...) outer product, silently mixing patients' masks	fixed; rank mismatches now raise
M3	ssim3d Gaussian window larger than the depth axis	wrong window on thin volumes; depth-1 now yields exact 2D SSIM	fixed (per-axis window, min(dim, window) forced odd)
M4	main.py scored the trainer's returned model, i.e. the final epoch , while --eval-only scored the selected best checkpoint	every 3D metrics.json was systematically pessimistic — v3_3d_gan test roi_pearson 0.419 (final) vs 0.482 (best). main2d had always reloaded correctly	fixed; main.py now honours --eval-ckpt and reloads before final eval
M5	main2d never wrote config.json or metrics.json	2D results existed only inside SLURM .err logs — not re-verifiable, and lost on log rotation	fixed; both written, metrics.json carries val and test plus the checkpoint used

6.3b Defects found while producing artifacts

#	defect	effect	status
A1	GAN decoder built from ConvTranspose3d (and the same in gan2d/flow2d)	period-2 checkerboard stamped into predictions: lag-1 autocorrelation of the first difference -0.60 (white noise is -0.50) against +0.81 for real DCE, 5× the target's HF power. Halved FID once removed (175 -> 89)	fixed — upsample+conv everywhere; guarded in selfcheck

#	defect	effect	status
A2	generate_synth2d restored predictions with uncrop_pad + CopyInformation	pure index arithmetic with no resample: correct only when the model grid and native grid share spacing. Under --reference iso a 1024 ² @0.176 mm case would have been written at half scale	fixed — uncrop onto the model grid, then resample to native
A3	generate_synth2d ran without no_grad	GAN path crashed on .numpy(); flow paths survived only because their samplers disable grad internally	fixed
A4	audit_baselines had never been executed	parse_args(argv) (takes no argv) and cond[i][0][m] (rank mismatch) — the identity baseline, the script's whole purpose, could never have run	fixed; now also reports per-case roi_pearson for model vs t2w
A5	train.slurm had no USE_PREGAD knob (train2d.slurm did)	USE_PREGAD=1 was silently ignored; an 8 h run trained 3-channel while believed to be 4-channel	fixed; the script now echoes an effective: line of the silent-failure-prone toggles
A6	SliceDCEDataset stacked via torch.stack	list and stacked tensor coexist, so peak RSS ~ 2× the cache (~68 GB); OOM-killed all four 2D runs at the end of slice-prep	fixed (drain-while-copying, ~1×); --mem 64G -> 96G

6.3c Leakage

#	defect	effect	status
L1	_build_data_ucsf fit the Nyul harmonizer over the entire cohort , test patients included	held-out intensity statistics fed the landmark fit that every split's normalization depends on. The Bao path had always excluded its test hospitals; the UCSF path never did	fixed — fits TRAIN indices only, via a ucsf_split_indices helper shared with the splitter so they cannot drift

L1 did not affect any run reported here: all UCSF runs pass --t2w-norm percentile --dce-norm robust, leaving nyul_modalities empty so the fit is skipped entirely (“harmonizer all per-image; no Nyul fit needed” in the logs). But train.slurm defaults T2W_NORM=nyul, so any run not overriding it would have leaked.

A full audit of the remaining leakage surface came back clean: train/test disjoint and exhaustive; val carved from train only; test_loader passed into the trainers but never used; LDM scaling_factor computed from train latents and recomputed identically on load; 2D splits patients *before* slicing (no slice-level leakage); evaluate and val_score apply identical clamping and share metric code.

M1 is the most important single finding in the audit: it is the reason earlier roi_pearson figures looked encouraging, and it invalidated the two conclusions previously drawn from them — that the **data-scale hypothesis was rejected** and that the **~0.5 localization ceiling was information-bound**. Both were measured on crippled configurations and are retracted; neither has been re-tested.

6.4 Infrastructure

--mem=32G OOM-killed on 70-phase series (-> 96G); /mnt/fac is absent (ENODEV) on some CPU nodes (-> preflight mount check, exit 17, exclude the bad node); train2d.slurm was missing the HF_HOME/HF_HUB_OFFLINE/TORCH_HOME exports that train.slurm has, which killed the MedVAE run on a HuggingFace fetch.

6.5 Regression guard

There is no unit-test suite. tier1_static/selfcheck.py provides 26 preflight assertions across metrics/loss, model/determinism and data/staging that run on CPU in seconds; it is the intended gate before launching cluster runs. Two of the bugs above (D5, and an earlier enh_max case) initially escaped because the **fixture did not mirror the real staged tree** — fixtures must be built from staged filenames, not idealized ones.

7. Validity of current results

7.1 Run-to-run variance — read this before quoting any comparison

Two **config-identical runs at the same seed** (v3_3d_gan and the accidental replicate v4_3d_gan_pregad, whose USE_PREGAD=1 was silently dropped, §A5) differ materially. set_seed seeds Python/NumPy/Torch but does **not** set cudnn.deterministic, cudnn.benchmark=False, or use_deterministic_algorithms — and 3D conv backward uses non-deterministic atomicAdd regardless. Adversarial training then amplifies ~1e-7 divergences, and best-checkpoint selection turns a bumpy val trajectory into a noisy *discrete* choice of epoch, which moves every metric at once.

metric	v3	replicate	delta
roi_pearson	0.4824	0.4657	-0.017
ssim_roi	0.3968	0.3887	-0.008
mae_roi	0.2355	0.2441	+0.009
roi_var_ratio	0.9123	0.8577	-0.055
roi_grad_ratio	0.7105	0.6763	-0.034
p75_corr	0.3570	0.0812	-0.276

This is one replicate pair, so it bounds the noise magnitude but is not a standard deviation; >=3 seeds are needed for that.

Why p75_corr is 16× worse than roi_pearson. roi_pearson is a within-patient statistic averaged over 550 patients, so averaging suppresses variance. p75_corr is a single cross-patient correlation, and it measures the *amplitude* dimension — which the audit (§7.2) shows the model barely predicts at all. Nothing in an ROI-weighted L1 on a homogeneous target pins down a global offset, so amplitude drifts

freely between runs. **Reproducibility tracks identifiability:** the well-determined component (structure) reproduces at ± 0.017 ; the ill-determined one (amplitude) does not.

Consequences. Differences below ~ 0.02 `roi_pearson` are unresolved at $n=1$. That retires one earlier claim: the 2D iso-vs-no-iso A/B differed by 0.020 on `roi_pearson`, i.e. within noise, so **iso cannot be said to improve metrics** — it remains justified on correctness grounds (§1.7) only. `p75_corr` should not be quoted from a single run. Comparisons that survive comfortably: GAN vs flow (0.113, $\sim 7\times$), model vs t2w baseline (0.525, $\sim 30\times$), PZ vs TZ (paired within run), best vs last checkpoint (0.063, $\sim 4\times$).

7.2 What the model has actually learned (audit, $n=550$ test)

predictor	MAE_roi	roi_r	knows
const (cohort ROI mean)	0.2510	—	nothing
t2w copy	0.3372	-0.043	identity baseline
GAN	0.2355	+0.482	trained generator
level (oracle brightness)	0.1851	—	correct per-case brightness, no structure

Structure is learned; amplitude is not. Copying T2w scores ~ 0 , so `roi_pearson` 0.482 is genuine localization, not anatomy passthrough. But the model tracks per-case brightness at only $r = 0.334$ and compresses between-case spread $2.4\times$ (target sd 0.231 $\rightarrow$ 0.098), and **51% of target ROI variance is between-case brightness** — so both models lose to the oracle-brightness baseline on MAE, and the flow is worse than a *constant* (-0.018) while still having real localization ($r = 0.369$).

Physically coherent: per-case amplitude depends on injection dose/rate, cardiac output, bolus timing and scanner gain, almost none of which is visible in T2w/DWI/ADC. Within- gland structure is. Note PI-RADS curve types (I/II/III) are **scale-invariant**, so the failing component may be largely irrelevant to the Tier-2 clinical endpoint.

7.3 Standing caveats

- **All runs to date train on the §1.7 FOV mixture** — an $8\times$ range of anatomical scale; for the 17% cropped below 80 mm, ROI metrics approach whole-image metrics. The GAN-vs-flow *ranking* holds (same mixture for both); absolute numbers are not a clean baseline. Runs from `v3_*` onward use `--reference iso` and are exempt.
- **3D metrics.json written before the M4 fix reports the final epoch, not the best checkpoint.** Re-run with `--eval-only` to compare like with like.
- 2D and 3D ROI metrics are **not** comparable: the 2D dataset keeps only gland-containing slices, so the denominators differ.
- Every result predating the audit is superseded.

8. Reproducing the current cohort

```
# stage once, with the QC filter inline (yields 4246)
MIN_ENH=1.5 sbatch --array=0-3 --exclude=gcpu2-17 tier1_static/scripts/stage_ucsf.slurm

# verify: expect 4246, median enh ~2.07, p10 1.70, no t=0 warning,
#         27 interleaved, 155 curve-reselected
python -m tier1_static.data.stage_ucsf --report
python -m tier1_static.selfcheck          # 26 assertions

# train (logs must show `min_b=600` and `562 below b600, 17 without enhancement QC`)
python -m tier1_static.main2d --model gan \
```

```
--ucsf-main-root <staged> --dwi-min-bvalue 600 --require-qc \  
--dce-target-time 120 --dce-norm robust --dce-robust-k 1 \  
--spatial-size 32 256 256 --roi-weight 10
```

Add `--reference iso --iso-spacing 0.35 0.35 3.0` to address §1.7. Validated against the cohort (no-op for the 97.8% at 3.00 mm and for the dominant in-plane group; no crop-center shift, since T2w and DCE grids are identical in all 4246 cases) but **not yet used by any trained model** — the next round of runs should adopt it, and it is the natural candidate to become the default for UCSF.